Week 3, Lecture 1: Probability

Instructor notes – Part 1: prep reading, Part 2: podium cheat sheet

This single file has two parts, both instructor-only (not a student handout). **Part 1** is read beforehand – full worked examples and the reasoning behind the flow. **Part 2** (after the page break, large print) is the podium copy.

Source material: Bolker 4.1-4.3 (intro through Bayes' Rule, p. 139-152), leaning on the false-positive medical testing example (4.3.1, p. 148) and the liana infestation example (4.3.2, p. 149). Also mined: `LECTURE2.Rmd` (the legacy, pre-redesign version of this exact lecture – its two-attribute urn example is well-tested and reused directly below) and the idea bank's PPV problem.

Where this sits in the week

- Today: probability foundations – sample space, events, axioms, joint/marginal/conditional probability, independence, Bayes' rule.
- Tomorrow (Lab 3): Act 1 applies today's Bayes' rule machinery to a camera-trap classifier PPV problem (different numbers, same technique); Acts 2-3 get hands-on with `dbinom()/dpois()/dnorm()` *before* Wednesday's lecture formally introduces them.
- Wednesday: the bestiary of distributions – picks up where today leaves off, now that students have both the theory (today) and hands-on exposure (lab) in hand.

Rationale for the flow

Bolker's own chapter opens with sample space/event/probability-measure formalism before getting to anything useful. Keep that section short – Calc-II students need the vocabulary but not a measure-theory detour. Spend the real time on **one worked example that carries joint, marginal, conditional probability, and independence all in one place** (the urn), then a **second worked example that carries Bayes' rule** (the medical test). Two examples, not five – depth over breadth, since tomorrow's lab is where students actually practice this.

Part A: Foundations (~8 min)

Open with an ecological framing, not bare notation: a site is either occupied by a species or not, before you go survey it – that's already a random variable in the making.

- **Sample space** Ω: the set of all possible outcomes of a trial. For two fair coin flips, $\Omega = \{HH, HT, TH, TT\}$.
- **Event**: a subset of Ω that we assign a probability to (e.g. "at least one head" = $\{HT, HH, TH\}$).
- **Random variable**: a mapping from Ω onto the real number line (e.g. $X = 0$ for tails, $X = 1$ for heads – the assignment is arbitrary, but it lets us do arithmetic with outcomes).
- **Kolmogorov's axioms** (all of probability theory follows from just these three):
  1. $P(E) \geq 0$ for any event E – probabilities are never negative.
  2. $P(\Omega) = 1$ – something in the sample space always happens.
  3. For disjoint events, $P(\bigcup_i E_i) = \sum_i P(E_i)$ – you can add up probabilities of non-overlapping pieces.

Sticking point to flag right away: students often want to treat "probability" as a property of an event in isolation. Axiom 3 is the one that does all the real work later (law of total probability, Bayes' rule) – flag that this is the one to remember.

Part B: Joint, marginal, conditional probability (the urn) (~10 min)

Reuse the two-attribute urn from the legacy lecture – it's compact and carries four concepts in one example. An urn contains objects that vary in both color and shape:

```
Urn <- matrix(NA, 2, 2, dimnames = list(Color = c("red", "blue"), Shape = c("sphere", "c
Urn["red", "sphere"]  <- 39
Urn["blue", "sphere"] <- 76
Urn["red", "cube"]    <- 101
Urn["blue", "cube"]   <- 25
Urn
```

```
##       Shape
## Color  sphere cube
##   red      39  101
##   blue     76   25
```

```
Prob <- Urn / sum(Urn)
Prob
```

```
##       Shape
## Color     sphere      cube
##   red  0.1618257 0.4190871
##   blue 0.3153527 0.1037344
```

Marginal probability – sum out the attribute you don't care about:

```
Prob_color <- rowSums(Urn) / sum(Urn)
Prob_shape <- colSums(Urn) / sum(Urn)
Prob_color
```

```
##       red      blue
## 0.5809129 0.4190871
```

```
Prob_shape
```

```
##    sphere      cube
## 0.4771784 0.5228216
```

Joint probability, done wrong first (on purpose): ask "what's $P(\text{blue} \cap \text{cube})$?" and let someone suggest $P(\text{blue}) \times P(\text{cube})$:

```
Prob_color["blue"] * Prob_shape["cube"]      # ~0.219 -- WRONG
```

```
##      blue
## 0.2191078
```

```
Prob["blue", "cube"]                         # ~0.104 -- actual value
```

```
## [1] 0.1037344
```

They don't match. That's the whole point: multiplying marginals only gives the joint probability **if the two attributes are independent**, and color and shape aren't independent in this urn (cubes are disproportionately red). This motivates conditional probability as the *correct* general tool:

$$P(\text{blue} \mid \text{cube}) = \frac{P(\text{blue} \cap \text{cube})}{P(\text{cube})}$$

```
Prob["blue", "cube"] / Prob_shape["cube"]
```

```
##      cube
## 0.1984127
```

Rearranging gives the general joint-probability formula (true whether or not the events are independent):

$$P(A \cap B) = P(A \mid B)\, P(B)$$

Independence, properly defined: A and B are independent exactly when $P(A \mid B) = P(A)$ – conditioning on B doesn't change your belief about A. Only in that special case does the joint probability collapse to the product of marginals. This is the fact underlying nearly every independence assumption made for the rest of the semester (iid data, independent trials in a binomial, etc.).

Part C: Bayes' rule (~12-14 min)

Derive it as "a recipe for turning around a conditional probability." Starting from the joint-probability formula from Part B, applied both ways:

$$P(A \cap B) = P(A \mid B)P(B) = P(B \mid A)P(A)$$

$$\Rightarrow \quad P(A \mid B) = \frac{P(B \mid A)\, P(A)}{P(B)}$$

Relabel with H (hypothesis) and D (data) – the framing we'll live in for the rest of the semester:

$$P(H \mid D) = \frac{P(D \mid H)\, P(H)}{P(D)}$$

$P(D \mid H)$ is the **likelihood** (the single most important quantity in this course, frequentist or Bayesian). $P(H)$ is the **prior**. $P(D)$ is the unconditional/marginal probability of the data – usually the hard part, computed via the law of total probability (axiom 3 again) over a set of exhaustive, mutually exclusive hypotheses.

Worked example – false positives in medical testing (Bolker 4.3.1, using his exact numbers so students can follow along in the reading):

```
P_I <- 1e-6          # P(infected) -- unconditional prevalence
P_pos_given_I <- 1    # sensitivity: never a false negative
P_pos_given_U <- 1e-2 # false positive rate

P_U <- 1 - P_I
P_pos <- P_pos_given_I * P_I + P_pos_given_U * P_U
P_pos                                      # ~= 1e-2 -- the 1e-6 term is negligible
```

```
## [1] 0.01000099
```

```
PPV <- (P_pos_given_I * P_I) / P_pos
PPV                                        # ~= 1e-4
```

```
## [1] 9.99901e-05
```

Walk through *why* $P(+) \approx 10^{-2}$: the true-positive contribution (10^{-6}) is four orders of magnitude smaller than the false-positive contribution ($\approx 10^{-2}$), so it can be dropped. That approximation move – neglecting a term four orders of magnitude smaller than another – is worth naming explicitly; it's the same "throw out the lower-order term" instinct from Week 2's limits, now applied to arithmetic instead of algebra.

Sticking point to flag – the single most common error here: confusing $P(H \mid D)$ with $P(D \mid H)$. The test has excellent sensitivity ($P(+ \mid I) = 1$) but that number answers a completely different question than the one anyone actually cares about ($P(I \mid +)$). This confusion has a name – the **base rate fallacy** – and it recurs constantly outside the classroom (DNA evidence in forensics is Bolker's other example, p. 149).

If time – second example, multi-hypothesis Bayes' rule (liana infestation, Bolker 4.3.2, p. 149): same idea generalized from 2 hypotheses to N: $P(H_i \mid D) = \dfrac{P(D \mid H_i)P(H_i)}{\sum_j P(D \mid H_j)P(H_j)}$. Just flash the setup (4 liana infestation classes, a new tree's number of infested neighbors as the data) – don't work it in full; tomorrow's Lab 3 Act 1 does a full worked version of this idea with a camera-trap classifier instead of lianas.

Take-home message

- Axioms 1-3 are the whole foundation; axiom 3 (summing disjoint pieces) is what powers the law of total probability and Bayes' rule.
- Joint probability is $P(A \mid B)P(B)$ in general; it only simplifies to $P(A)P(B)$ when A and B are independent – don't assume independence, check it.
- Bayes' rule turns $P(D \mid H)$ (what you can compute) into $P(H \mid D)$ (what you actually want) – and a "good-sounding" test can still have a surprisingly low $P(H \mid D)$ when $P(H)$ is small.

Roadmap tie-in

- Tomorrow (Lab 3, Act 1): same Bayes' rule machinery, camera-trap classifier PPV, different numbers.
- Wednesday: today's foundations become the platform for the bestiary of named distributions – discrete and continuous "shapes" for describing variability.

Discussion seed (if time remains)

Ask: *if a camera-trap classifier flags an image as your target species, and you know the species is rare at this site, should you trust a single flag? What would change your answer – a second, independent flag? A different kind of evidence entirely?* Good prompt for previewing tomorrow's lab and for surfacing that "how much evidence is enough" is itself a probability question.

PART 2 – PODIUM CHEAT SHEET

Podium copy – terse on purpose. Full derivations, rationale, and code: see Part 1 above.

0. Open (~2 min)

- Today: probability foundations. Tomorrow's lab and Wednesday's lecture both build directly on this.

1. Foundations (~8 min)

- Sample space Ω, event (subset of Ω), random variable (mapping onto $\mathbb{R}$)
- **Kolmogorov's axioms:**
  1. $P(E) \geq 0$
  2. $P(\Omega) = 1$
  3. disjoint events: $P(\cup E_i) = \sum P(E_i)$ – **this one does all the work later**

2. Urn example: joint/marginal/conditional/independence (~10 min)

- Marginal: sum out the attribute you don't want (`rowSums`/`colSums`)
- **Wrong-then-right joint probability:** $P(\text{blue})P(\text{cube}) \approx 0.219$ vs. actual $P(\text{blue}, \text{cube}) \approx 0.104$ – multiplying marginals only works under independence!
- Conditional: $P(A \mid B) = P(A \cap B)/P(B)$
- General joint formula: $P(A \cap B) = P(A \mid B)P(B)$
- **Independence** (proper def.): $P(A \mid B) = P(A)$

3. Bayes' rule (~12-14 min)

- From $P(A \cap B)$ both ways: $P(H \mid D) = \dfrac{P(D \mid H)P(H)}{P(D)}$
- $P(D \mid H) =$ **likelihood**, $P(H) =$ **prior**, $P(D) =$ law of total probability over exhaustive hypotheses

Medical test (Bolker 4.3.1): $P(I) = 10^{-6}$, sensitivity $= 1$, false positive rate $= 10^{-2}$

- $P(+) \approx 10^{-2}$ (the 10^{-6} term is negligible – same "drop the small term" move as Week 2's limits)
- PPV $= P(I \mid +) \approx 10^{-4}$ – a "good" test, still a surprisingly unreliable positive result
- **Sticking point:** $P(I \mid +) \neq P(+ \mid I)$ – the **base rate fallacy**

If time: liana infestation (Bolker 4.3.2) – same idea, N hypotheses instead of 2. Flash the setup only; full version tomorrow.

4. Roadmap + discussion (~3 min)

- Tomorrow: Lab 3 Act 1, same machinery, camera-trap classifier
- Wednesday: bestiary of distributions
- *Discussion:* would a second, independent flag change your trust in a positive detection? What kind of evidence would?